CSE 270 Sports Analytics

Rasch model

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Bradley-Terry and Rasch model

- In Bradley-Terry model each team is either home team or away team
- We look at how many times a team being home team one against the other team being away team, and vice versa
- Bradley-Terry helps to rank teams by their ability to win

Bradley-Terry and Rasch model

- In Rasch model every team is either on defense or on offense.
  - For each possession during the game, there are 5 offensive and 5 defensive
- If Bradley-Terry model helps to rank the teams, Rasch model helps to estimate the players offensive and defensive skills

Rasch model for NBA

$$E[P_i] = \alpha + \sum_{i=1}^{5} \beta_{O_{ij}} + \sum_{i=1}^{5} \delta_{D_{ij}} + \theta H_i$$

Where,

i = 1:n, *number of possessions*, j = 1:5

- O_{ij} identifies j^{th} offensive player on i^{th} *possession*
- D_{ij} identifies j^{th} defensive player on i^{th} *possession*
- H_i indicates whether offense is the home team on i^{th} *possession*
- P_i is the number of points scored on i^{th} *possession*
- $E[P_i]$ is the expected number of the points scored on i^{th} *possession*

Rasch Model

$E[P_i] = \alpha + \sum_{i=1}^{5} \beta_{O_{ij}} + \sum_{i=1}^{5} \delta_{D_{ij}} + \theta H_i$

Estimated β_{O_j} coefficient will show the jth player's offensive strength

δ_{D_j} coefficient will show the j^{th}player's defensive strength

The matrix representation of the model

Independent variables

$$\mathbf{X} = \overbrace{\begin{pmatrix} X_{11} & \dots & X_{1p} \\ X_{21} & \dots & X_{2p} \\ X_{31} & \dots & X_{3p} \\ \dots & \dots & \dots \\ X_{(n-1)1} & \dots & X_{(n-1)p} \\ X_{n1} & \dots & X_{np} \end{pmatrix}}^{\text{Offense}} \overbrace{\begin{pmatrix} X_{1(p+1)} & \dots & X_{1(2p)} \\ X_{2(p+1)} & \dots & X_{2(2p)} \\ X_{3(p+1)} & \dots & X_{3(2p)} \\ \dots & \dots & \dots \\ X_{(n-1)(p+1)} & \dots & X_{(n-1)(2p)} \\ X_{n(p+1)} & \dots & X_{n(2p)} \end{pmatrix}}^{\text{Defense}} \begin{pmatrix} H_1 \\ H_2 \\ H_3 \\ \dots \\ H_{n-1} \\ H_n \end{pmatrix}_{n \times (2p+1)}$$

for $j = 1:5$

$$X_{ij} = \begin{cases} 1 & \text{if player } j \text{ is on } \textbf{offense} \text{ on possession } i \\ 0 & \text{otherwise} \end{cases}$$

for $j = p+1:2p$

$$X_{ij} = \begin{cases} 1 & \text{if player } j-p \text{ is on } \textbf{defense} \text{ on possession } i \\ 0 & \text{otherwise} \end{cases}$$

The matrix representation of the model

Dependent variable and the coefficients

$$\mathbf{y} = \begin{pmatrix} P_1 \\ P_2 \\ P_3 \\ \dots \\ P_{n-1} \\ P_n \end{pmatrix}_{n \times 1} \qquad \beta = \begin{pmatrix} \beta_1 \\ \dots \\ \beta_p \\ \delta_1 \\ \dots \\ \delta_p \\ \theta \end{pmatrix}_{p \times 1}$$

The data

The game Boston Celtics vs. New York Knicks - https://goo.gl/KSQBXe

```
load("./data/nba11_12.rda")
head(nba11_12)
```

```
##                 O1                 O2              O3            O4            O5
## 1   Kevin Garnett    Sasha Pavlovic Jermaine O'Neal     Ray Allen   Rajon Rondo
## 2 Carmelo Anthony Amar'e Stoudemire  Tyson Chandler Toney Douglas Landry Fields
## 3   Kevin Garnett    Sasha Pavlovic Jermaine O'Neal     Ray Allen   Rajon Rondo
## 4 Carmelo Anthony Amar'e Stoudemire  Tyson Chandler Toney Douglas Landry Fields
## 5   Kevin Garnett    Sasha Pavlovic Jermaine O'Neal     Ray Allen   Rajon Rondo
## 6 Carmelo Anthony Amar'e Stoudemire  Tyson Chandler Toney Douglas Landry Fields
##                 D1                 D2              D3            D4            D5
## 1 Carmelo Anthony Amar'e Stoudemire  Tyson Chandler Toney Douglas Landry Fields
## 2   Kevin Garnett    Sasha Pavlovic Jermaine O'Neal     Ray Allen   Rajon Rondo
## 3 Carmelo Anthony Amar'e Stoudemire  Tyson Chandler Toney Douglas Landry Fields
## 4   Kevin Garnett    Sasha Pavlovic Jermaine O'Neal     Ray Allen   Rajon Rondo
## 5 Carmelo Anthony Amar'e Stoudemire  Tyson Chandler Toney Douglas Landry Fields
## 6   Kevin Garnett    Sasha Pavlovic Jermaine O'Neal     Ray Allen   Rajon Rondo
##   Home Points       Date
## 1    0      0 2011-12-25
## 2    1      0 2011-12-25
## 3    0      2 2011-12-25
## 4    1      3 2011-12-25
## 5    0      0 2011-12-25
```

The data preperation

How many players are there ?

```
player <- unlist(nba11_12[, 1:10])
player[1:30]
```

```
##             O11             O12             O13             O14             O15
##   Kevin Garnett Carmelo Anthony   Kevin Garnett Carmelo Anthony   Kevin Garnett
##             O16             O17             O18             O19            O110
## Carmelo Anthony   Kevin Garnett Carmelo Anthony   Kevin Garnett Carmelo Anthony
##            O111            O112            O113            O114            O115
##   Kevin Garnett Carmelo Anthony   Kevin Garnett Carmelo Anthony   Kevin Garnett
##            O116            O117            O118            O119            O120
## Carmelo Anthony   Kevin Garnett Carmelo Anthony   Kevin Garnett Carmelo Anthony
##            O121            O122            O123            O124            O125
## Carmelo Anthony   Kevin Garnett Carmelo Anthony       Ray Allen Carmelo Anthony
##            O126            O127            O128            O129            O130
##       Ray Allen Carmelo Anthony Carmelo Anthony       Ray Allen Carmelo Anthony
## 478 Levels: A.J. Price Aaron Gray Al-Farouq Aminu Al Harrington ... Hamady Ndiaye
```

```
length(player)
```

```
## [1] 2097810
```

The data preperation

```
num.p <- length(unique(player))
num.p
```

```
## [1] 478
```

```
prefix = rep(c('off', 'def'), each = 5*nrow(nba11_12))
length(prefix)
```

```
## [1] 2097810
```

```
tag = paste(prefix, player)
length(tag)
```

```
## [1] 2097810
```

These will be our columns

The data preperation

```
tag[1:50]
```

```
##  [1] "off Kevin Garnett"   "off Carmelo Anthony" "off Kevin Garnett"
##  [4] "off Carmelo Anthony" "off Kevin Garnett"   "off Carmelo Anthony"
##  [7] "off Kevin Garnett"   "off Carmelo Anthony" "off Kevin Garnett"
## [10] "off Carmelo Anthony" "off Kevin Garnett"   "off Carmelo Anthony"
## [13] "off Kevin Garnett"   "off Carmelo Anthony" "off Kevin Garnett"
## [16] "off Carmelo Anthony" "off Kevin Garnett"   "off Carmelo Anthony"
## [19] "off Kevin Garnett"   "off Carmelo Anthony" "off Carmelo Anthony"
## [22] "off Kevin Garnett"   "off Carmelo Anthony" "off Ray Allen"
## [25] "off Carmelo Anthony" "off Ray Allen"       "off Carmelo Anthony"
## [28] "off Carmelo Anthony" "off Ray Allen"       "off Carmelo Anthony"
## [31] "off Carmelo Anthony" "off Rajon Rondo"     "off Carmelo Anthony"
## [34] "off Rajon Rondo"     "off Carmelo Anthony" "off Marquis Daniels"
## [37] "off Marquis Daniels" "off Carmelo Anthony" "off Marquis Daniels"
## [40] "off Carmelo Anthony" "off Marquis Daniels" "off Carmelo Anthony"
## [43] "off Marquis Daniels" "off Carmelo Anthony" "off Marquis Daniels"
## [46] "off Carmelo Anthony" "off Marquis Daniels" "off Carmelo Anthony"
## [49] "off Marquis Daniels" "off Carmelo Anthony"
```

The data preperation

What will be the dimensions of our matrix? Number of the columns:

478 players * 2 + 1 (for Home team indicator)= 957 Number of the rows:

209,781

Total number of entries: 957*209,781= 200,760,417

The data preperation

Making the Matrix

```
# Define the row and column of X in which to put each 1.
tag.factor = as.factor(tag)
i <- rep(1:nrow(nba11_12), 10)
j <- as.numeric(tag.factor)

# Construct X as a sparse matrix.
library(Matrix)
X <- sparseMatrix(i, j)

# Update i and j to include a column for home court advantage.

i.new <- c(which(nba11_12$Home == 1), i)
j.new <- c(rep(1, sum(nba11_12$Home)), j + 1)

# Construct X as a sparse matrix with the updated i and j.

X <- sparseMatrix(i.new, j.new)

y <- nba11_12$Points
```

The data preperation

Use the Sparse Matrix.

In numerical analysis and computer science, a sparse matrix is a matrix in which most of the elements are zero.

In our matrix ~99% of the entries are zeros, rest are one's. Sparse Matrix representation helps to save computing time.

Modeling and the coefficients

Regularized linear model with the package glmnet

```
library(glmnet)
```

```
## Loaded glmnet 4.1-4
```

```
set.seed(1)
model<- cv.glmnet(X, y, alpha = 0, standardize = FALSE)

coef <- coef(model, s = 'lambda.min')[, 1]
names(coef) <- c('Intercept', 'HFA', sort(unique(tag)))
```

Extracting coefficients

Extracting coefficients

```
# Intercept
alpha <- coef[1]
# Home Advantage
theta <- coef[2]
# Defensive strenght
delta <- coef[2 + 1:num.p]
# offensive strengh
beta <- coef[2 + num.p + 1:num.p]

length(delta)
```

```
## [1] 478
```

```
length(beta)
```

```
## [1] 478
```

Beta coefficients

Beta coefficients measure attacking skills of the player: Larger is the number, stronger is the player on offense

Top Ten offensive players

```
head(sort(beta, decreasing=T),n=10)
```

```
##      off James Harden       off LeBron James      off Dirk Nowitzki
##            0.04893444             0.03834923             0.03551613
##        off Chris Paul     off Blake Griffin     off Manu Ginobili
##            0.03074925             0.03038170             0.02980103
##       off Mike Miller off Russell Westbrook      off Andre Miller
##            0.02918340             0.02915810             0.02722582
##     off Ryan Anderson
##            0.02712350
```

Delta coefficients

Delta coefficients measure defensive skills of the player, smaller is the number (negative) better is the player on defense

Top 10 defensive players

```
head(sort(delta) , n=10)
```

```
##      def Taj Gibson   def Kevin Garnett  def Andre Iguodala     def Ricky Rubio
##         -0.04021855         -0.03304900         -0.02833276         -0.02826440
##      def Tim Duncan      def Marc Gasol def Thabo Sefolosha     def Ian Mahinmi
##         -0.02711243         -0.02642119         -0.02580529         -0.02437742
##  def Gerald Wallace      def Tony Allen
##         -0.02402187         -0.02399266
```

Making predictions

- Miami Heat vs. Dallas Mavericks.
- Miami Heat is on Offense
- The game is at Dallas Mavericks Court The following formation I on the court

Heat	O1	O2	O3	O4	O5
	LeBron James	Norris Cole	Udonis Haslem	Juwan Howard	Shane Battier
Mavericks	D1	D2	D3	D4	D5
	Jason Terry	Shawn Marion	Delonte West	Brian Cardinal	Brandan Wright

Predicted points for the offense team

Lets predict the points

$$E[P_i] = \alpha + \sum_{i=1}^{5} \beta_{O_{ij}} + \sum_{i=1}^{5} \delta_{D_{ij}} + \theta H_i$$

Heat	O1	O2	O3	O4	O5
	LeBron James	Norris Cole	Udonis Haslem	Juwan Howard	Shane Battier
Mavericks	D1	D2	D3	D4	D5
	Jason Terry	Shawn Marion	Delonte West	Brian Cardinal	Brandan Wright

$$P = \alpha + \beta_{\text{James}} + \beta_{\text{Cole}} + \beta_{\text{Haslem}} + \beta_{\text{Howard}} + \beta_{\text{Battier}} +$$
$$\delta_{\text{Terry}} + \delta_{\text{Marion}} + \delta_{\text{West}} + \delta_{\text{Cardinal}} + \delta_{\text{West}} + \delta_{\text{Cardinal}} + \delta_{\text{Wright}} + \theta * 0$$

Getting the coefficients

Offense

```
off<-c("off LeBron James","off Norris Cole","off Udonis Haslem",
       "off Juwan Howard","off Shane Battier")
beta[off]
```

```
##  off LeBron James   off Norris Cole off Udonis Haslem  off Juwan Howard
##        0.03834923       -0.01595075       -0.01469766       -0.01213861
## off Shane Battier
##        0.01199970
```

Defense

```
def<-c("def Jason Terry","def Shawn Marion", "def Delonte West",
       "def Brian Cardinal","def Brandan Wright"
)
delta[def]
```

```
##    def Jason Terry   def Shawn Marion   def Delonte West def Brian Cardinal
##        0.004897191        0.003625852       -0.016922591       -0.002185638
## def Brandan Wright
##        0.004074041
```

Getting the coefficients/predicting

```
alpha
```

```
## Intercept
## 0.9528359
```

```
theta
```

```
##        HFA
## 0.01861821
```

Expected Points

```
alpha+sum(beta[off])+sum(delta[def])+theta*0
```

```
## Intercept
## 0.9538867
```

On average we expect this formation to score 0.95 points per possession

Defensive and Offensive ratings

To calculate defensive and offensive ratings, we take theta/2 (suggested by Sam Powers) to assume a "neutral field" for the player

```
def.rating = 100*(alpha + theta/2 + delta)
off.rating = 100*(alpha + theta/2 + beta)
```

Who were the best defensive players in 2011-12? If you were to put each of them in a lineup with 4 average players, how many points per 100 possessions would that team allow on defense against a lineup of 5 average players?

```
head(sort(def.rating), n=10)
```

```
##      def Taj Gibson   def Kevin Garnett  def Andre Iguodala     def Ricky Rubio
##            92.19264           92.90960           93.38122           93.38806
##      def Tim Duncan      def Marc Gasol def Thabo Sefolosha     def Ian Mahinmi
##            93.50326           93.57238           93.63397           93.77676
##  def Gerald Wallace      def Tony Allen
##            93.81231           93.81523
```

Defensive and Offensive ratings

Who were the best offensive players in 2011-12? If you were to put each of them in a lineup with 4 average players, how many points per 100 possesions would that team score on offense against a lineup of 5 average players?

```
head(sort(off.rating, decreasing=T), n=10)
```

```
##      off James Harden       off LeBron James      off Dirk Nowitzki
##             101.10794              100.04942              99.76611
##        off Chris Paul      off Blake Griffin      off Manu Ginobili
##              99.28942              99.25267              99.19460
##       off Mike Miller off Russell Westbrook       off Andre Miller
##              99.13284              99.13031              98.93708
##     off Ryan Anderson
##              98.92685
```

Actual performance on defense and offense

We can also calculate the actual points scored allowed per 100 posessions

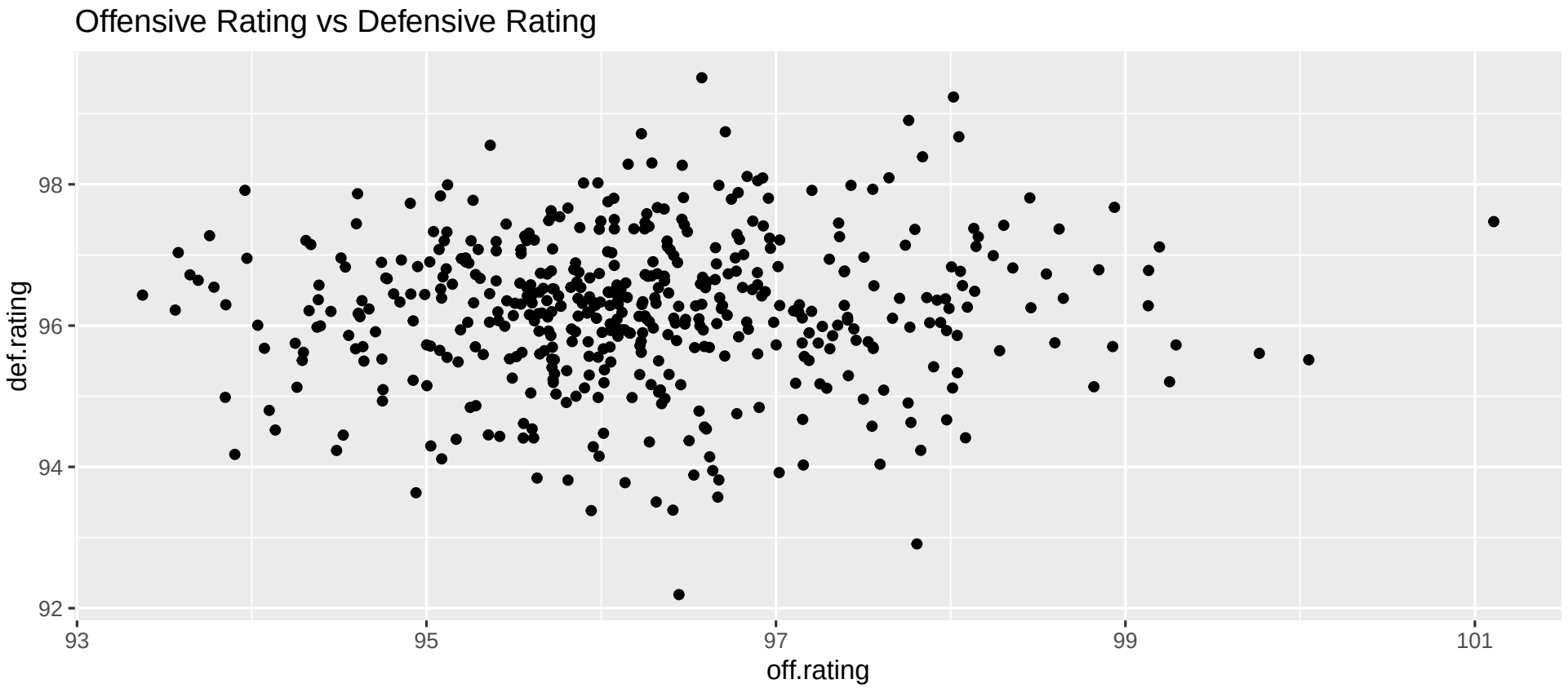
```
pp100 = 100*(crossprod(X, y)/colSums(X))[, 1]
names(pp100) = c('Home', sort(unique(tag)))
def.pp100 = pp100[1 + 1:num.p]
off.pp100 = pp100[1 + num.p + 1:num.p]
head(sort(off.pp100, decreasing=T), n=10)
```

```
##       off Ike Diogu   off Hamady Ndiaye     off Patty Mills   off Jerome Jordan
##            139.2857            114.2857            110.8108            107.4074
##     off Eric Maynor    off James Harden   off Nick Collison   off Manu Ginobili
##            107.1942            107.1523            107.0741            107.0286
##  off James Anderson off Jordan Hamilton
##            106.9121            106.7496
```

Offensive vs. Defensive rating

Is there a correlation between offensive and defensive ratings ?

```
ggplot()+geom_point(aes(x=off.rating, y=def.rating))+
  ggtitle("Offensive Rating vs Defensive Rating")
cor(off.rating, def.rating)
```

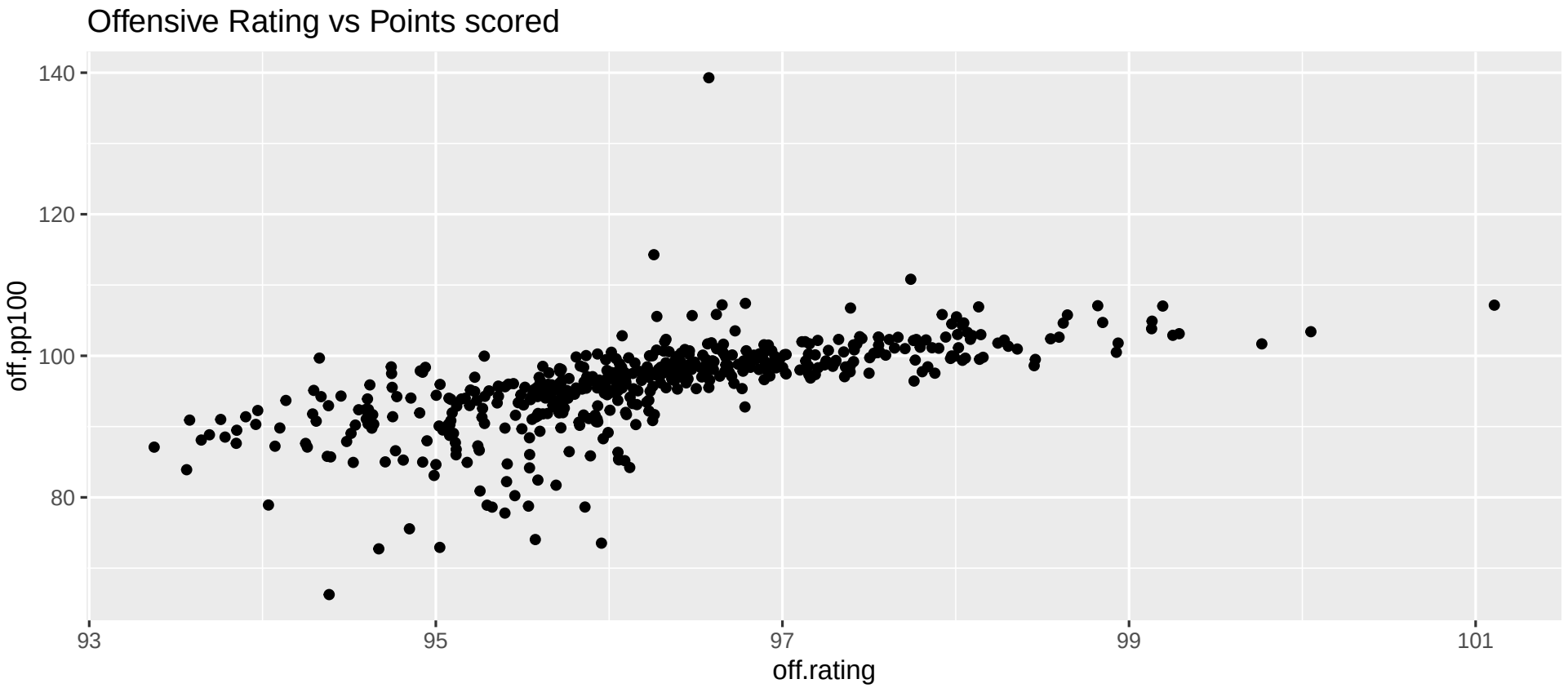



```
## [1] 0.04608928
```

Offensive rating vs actual performance

Is there a correlation between off rating and actual points scored

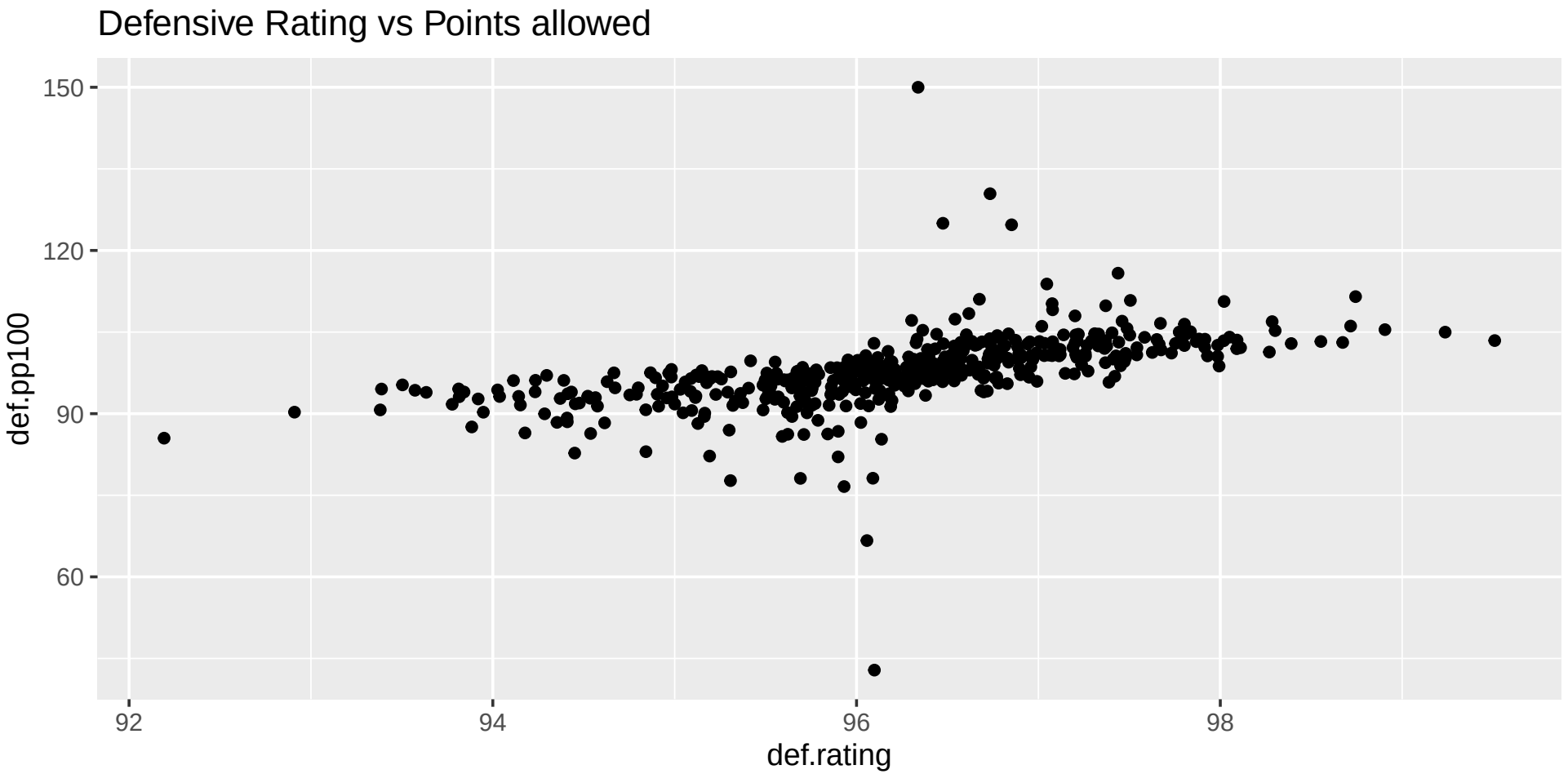
```
ggplot()+geom_point(aes(x=off.rating, y=off.pp100))+
  ggtitle("Offensive Rating vs Points scored")
cor(off.rating, off.pp100)
```




```
## [1] 0.6500404
```

Defensive rating vs actual performance

```
ggplot()+geom_point(aes(x=def.rating, y=def.pp100))+
  ggtitle("Defensive Rating vs Points allowed")
cor(def.rating, def.pp100)
```




```
## [1] 0.5430252
```